auxiliary reservoir and control reservoir volumes or brake pipe and brake cylinder volumes would indicate leakage into the brake pipe, presumably from the cylinder check valve in the quick action cap of the control valve, if the engine was alone. If coupled to a train it indicates back leakage from triple valves in the train.

Leakage from the straight air brake safety valve is either from dirt on the seat of its piston valve or from an improperly adjusted or defective reducing valve.

To determine which part is at fault the straight air brake can be applied and if the gauge registers 40 lbs. brake cylinder pressure it points to a leaky safety valve, but if the gange registers 50 lbs. or more (the figure of safety valve adjustment), it points to a defective or improperly adjusted reducing valve.

Wrongly Used Brake Valve.

During the past few months a number of our corresponednts have written us with reference to the effect of using the H 5 and H 6 brake valves on the wrong equipment, that is, through some mistake the H 5 brake valve would be used on the No. 6 equipment or the H 6 brake valve would be used on the No. 5 equipment.

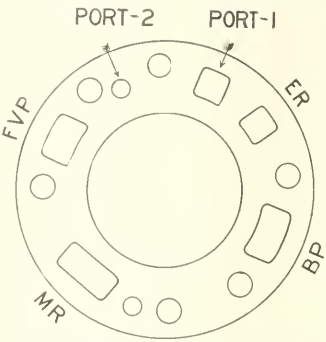
Our readers are well aware that the H 6 brake valve can be distinguished from the H 5 valve by the brass name plate attached to the rotary valve seat of the H 6 brake valve, but as the screws holding the plate are sometimes removed and used for other purposes, and the screw holes become filled with dirt, the H 6 valve is sometimes mistaken for the H 5. Should the name plate of the H 6 valve be removed, this valve can still be distinguished from the H 5, without taking the valve apart as a photographic view of the H 6 valve taken from a point facing the service position handle notch of the upper case, will show two series of lugs or projections on the upper case, rotary valve seat, and lower case, one series on each side of the valve when facing it from the point mentioned. Those lugs are intended to facilitate the operation of separating the different portions of the brake valve and a similar view of the H 5 brake valve shows when facing the service position notch on the upper case, but one series of lugs cast on the portions mentioned, and very near the point of the location of the number plate on the H 6 valve there is a small raised portion on the H 5 valve and by this the brake valves can be distinguished at a glance.

The reason that the valves resemble each other so much in size and appearance is that the valve handle, rotary key, upper case and lower case, as well as all the gaskets, are a standard for both brake valves, the only difference in the two valves being in the ports and passages in the rotary valve and seat, therefore either brake valve fits either pipe bracket.

In order to determine the effect of using the wrong brake valve on one of these equipments, it is only necessary to consider the brake valve, the bolt holes and port openings are the same in both middle and lower gaskets the only difference being that the upper gasket is made of leather and the lower of rubber and to find the effect of the wrongly used brake valve it is only necessary to consider the port openings marked 1 and 2 in the sketch of the port openings through the body gaskets.

The port openings in the lower case and the base of the rotary valve seat must, of course, register with the openings through the gaskets and on a No. 6 equipment, port shown as No. 1 leads to the release pipe and exhaust port of the equalizing slide valve and port No. 2 leads to a branch pipe to the application cylinder pipe.

On the No. 5 equipment port No. 1 leads to the application chamber pipe and port No. 2 to the branch of the double



VIEW OF PORTS AND BOLT HOLES FROM THE UPPER SIDE OF THE LOWER CASE OF H5 OR H6 BRAKE VALVE.

heading pipe between the brake valve and the cut-out-cock.

As the admission of main reservoir pressure, the equalizing discharge feature and the exhaust of brake pipe pressure is identical in both brake valves, the same ports through the lower case conduct the pressure to and from the brake system regardless as to whether the upper portion is of the H 5 or H 6 brake valve, but the two numbered ports are used for different purposes by the two valves.

Knowing where these ports lead to we will assume that the No. 6 equipment has in some way had an H 5 brake valve applied to it and we wish to know its effect.

It is manifest that the brakes will apply and release in the same way that the H 6 brake valve would apply or release them, but when the valve handle of the H 5 brake valve was returned to lap position the application cylinder pressure would escape and the engine brake would release because the H 5 brake valve opens the double heading pipe to the atmosphere in this position and on the No. 6 equipment a branch of the application cylinder pipe connects with this open port at the pipe bracket.

Should this branch of the application cylinder pipe be plugged or should it contain a check valve, as it does when the brake valve cut-out-cock is located in the reservoir pipe, the H 5 valve will do the same work that the H 6 valve will, with the single exception of maintaining the application cylinder pressure when the valve handle is in emergency position. Locating the brake valve cut-out-cock in the reservoir pipe is a Pennsylvania R. R. design, but is sometimes used on other systems.

Should the H 6 brake valve be used on the No. 5 equipment no difference in the action of the brake would be noticed in any position of the brake valve while this engine is handling the brakes in a train.

In emergency position there would be no flow of pressure to the distributing valve from the brake valve, but the pressure maintaining feature of the No. 5 equipment is made in the distributing valve. The H 6 valve would also exhaust the equalizing reservoir pressure instead of admitting it to the application chamber, but those features are of minor import when the exhaust and restoration of brake pipe pressure is the chief issue.

While the brake valves may become mixed and wrongly used and act in the manner described it is needless to say that in all cases the brake valve should be used with the equipment for which it is intended.

It is not difficult to distinguish any parts of the No. 5 equipment from those of the No. 6 as the brake valves and distributing valve of the No. 6 equipment have the name plate and should the plates be removed or accidentally torn off, the automatic brake valves can be distinguished as stated, and while the other valves of the equipments are similar in size and outward appearance, the S F 1 brake valve of the No. 5 equipment has 3 pipe connections while the S 6 valve has 4.

The distributing valve reservoir of the No. 5 equipment has a 3/8-inch brake pipe connection while the reservoir of the No. 6 brake has a 3/4-inch brake pipe connection.

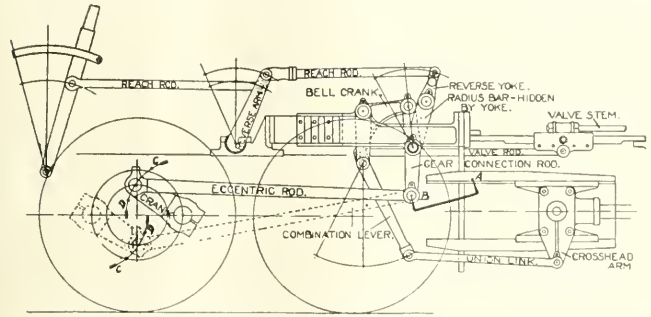
The E 1 safety valve connects with the No. 5 distributing valve by means of a standard 1/2-inch pipe thread, whereas the E 6 safety valve is screwed into the No. 6 distributing valve through a threaded connection, that is 17/16 of an inch in diameter and 16 threads per inch.

It is a good and safe rule to sojourn in every place as if you meant to spend your life there, never omitting an opportunity of doing a kindness, or speaking a true word, or making a friend.—*John Ruskin.*

General Foremen's Department

The Baker Locomotive Valve Gear.

The interest which has always attached to the subject of valve gearing seems perennial, and recent years bringing as they have an increase in the size of locomotives, naturally and properly brought a new interest to the subject of valve gearing. The limit of space inside the engine frames doubtless called the Walschaerts gearing into American locomotive service. In the important features of accessibility it is certainly an improvement on the so-called Stephenson gearing. But American enterprise did not stop at the mere adoption of the clever Belgian device. Hence the appearance some years ago of the Baker-Pilliod valve gear, which after passing through the experimental stage, and undergoing several organic changes, has now, with increased merit in point of efficiency, taken a settled form and is now known as the Baker valve gear.



THE BAKER VALVE GEAR.

Over 400 American locomotives are now equipped with this valve gearing and nearly 50 railroads have installed the gearing as part of their equipment and the reports, from all who have reported, are of the most gratifying kind.

A brief description of the gearing will be of interest, as the variation from any other form of valve gear is particularly marked. Like the Walschaerts gearing the movement is derived from the crosshead and a single eccentric crank. The crosshead connection has the effect of moving the valve the amount of the lap and lead, and the eccentric crank gives the required movement for the amount of valve travel. These peculiarities are similar to the movement in the Walschaerts valve gear, but whereas the Walschaerts gear is equipped with an oscillating link through which the movement passes to the radius bar and valve rod, the designers of the Baker gearing have dispensed with the use of the link and by an ingenious bell crank connection the motion is transferred to the radius bar and valve rod. As is well known among engineers, the avoidance of the variations due to the slip of the link and link block is a decided gain in valve wear mechanism, especially when the parts gather lost motion incidental to service.

In this regard the Baker valve gear lends itself readily to massiveness of construction. The bearings and bushings may readily be made of ample dimensions, and a degree of security and an assurance of a continuity of fixity in position may be attained hitherto unknown in valve gear construction and adjustment.

In addition to this important advantage there are other peculiarities in the movement which have been cleverly used to advantage by the inventors. It will be readily noted that when the piston is at the end of the stroke and the valve opening begins the eccentric rod is then passing through space at its highest point of speed. At the same time the crosshead connection has the effect of further accelerating the motion of the valve which opens the port rapidly, and as the crosshead advances toward the middle of the stroke, increasing in its velocity, the eccentric rod diminishes in speed and eventually lingers at the point where the steam port is completely open, and as the point of cut-off approaches the valve is closed with an increasing degree of rapidity. This eccentricity of movement is more or less exemplified in almost all kinds of valve gearing where steam is used, but in none is it more pronounced than in the case of the Baker gearing, with the exception, perhaps, of the Corliss valve gear, which with its delicate mechanism is not adapted for use on locomotive service

The readiness with which the Baker gearing may be applied to any locomotive is of advantage in its introduction on locomotives now in use. Engines equipped with the Stephenson valve gear where the parts are much worn may have the Baker gear applied at less cost than is necessary for a reconstruction of the older gearing. While many new locomotives are being equipped with the new gear, of which the Mikado type of locomotive, furnished by the Baldwin Locomotive Works, and which are described in another part of the current issue of our magazine, are an illustration, many locomotives that have seen much service are being refitted with the Baker valve gearing, and the advantage gained is, as far as we have been able to learn, very warmly appreciated.

Roundhouse Efficiency.

By T. F. Griffin, General Foreman, C. C. C. and St. L. Ry., Indianapolis, Ind.

Regarding roundhouse efficiency will state that the first thing to do is to get our engines in the roundhouse as soon as possible after arriving, so we can get to work on them and get them ready for service. To take note of engine crews coming in with heavy bad fires that take twice as long to get fire knocked out as other engines of the same class and service, to make report of this to road foreman. To get after crews that they are delaying engine and wasting coal, as unburnt coal goes in cinder car.

To provide a clean, comfortable washroom with hot and cold water for engine crew, close to pit so crew can go there and wash, and not wash on engine and hold the engine until they get washed and dressed, and tell all about their trip, as this will delay engine from 15 to 30 minutes. To have a live man for hostler and fire knocker, a good ash pit well drained, a good coal chute to hold 200 to 400 tons, take coal and sand on both sides, use one side for incoming and other side for out-going track. Have an active roundhouse foreman and boiler foreman, so they know as soon as engine strikes turntable from ash pit, if engine is due for a washout or not, so they will start cooling them down the minute engine arrives at roundhouse. Have a good inspector of engine, tank and air. Inspector to read engineer's work report on work book, then make a thorough inspection, make a report of all defects that he finds in writing and give to roundhouse foreman, to start the different class of men to work on engines,

Catechism of Railroad Operation

By Angus Sinclair

QUESTIONS AND ANSWERS.

Third Series Continued.

152. What should be done in case a reach rod breaks?

A. Block should be placed over the link block in the position that will enable the engine to start the train and to keep it moving slowly.

153. In case the reverse lever or reach rod got jammed at short cut-off, what should be done?

A. It would be necessary to remove the pin securing the reach rod to the tumbling shaft and then to take the same measures as with a broken reach rod.

154. For what breakages is it necessary to take down the main rod?

A. Broken piston, cross-head or joints, broken valve rod when steam cannot be kept out of the cylinder, broken main rod, strap or crank pin.

155. What accidents call for side rods to be taken down?

A. Broken side rod, broken main pin or broken side rod pin affecting that side.

156. When an accident happens that calls for the covering of the steam ports without the main rod being removed, how can the piston be oiled?

A. Sometimes it is possible to block the valve in such a way that oil will pass into the cylinder from the lubricating pipe. If this cannot be done oil must be squirted into the cylinder through some of the openings, such as can be made by removing relief valves or indicator plugs. When this cannot be done successfully it may be necessary to slack the cylinder head.

157. What would you do if the main rod back brass got too hot for safe running?

A. Would stop and cool off the pin, slack the key, administer a good supply of grease and graphite, then proceed slowly.

158. How would you proceed if the whistle stem broke?

A. Would keep injectors going till steam ceased to work them. Then would remove the broken part from the dome and drive a hard wood plug into the hole.

ACCIDENTS TO RUNNING GEAR.

159. What should be done in case a driving axle broke?

A. With a main driving axle broken close to the wheel all rods on the disabled side must come down. With any other driving axle broken, only such rods should come down as would give trouble to the other rods. With a broken main axle the cross-head must be blocked after the main rod has been taken down and the valve secured in the middle of its seat, the valve stem being disconnected. Before the rods have been taken down the broken end of the axle should be jacked up so that a good solid block of wood can be put in place of the oil cellar on that side, then the space between that block and the pedestal brace must be filled by another solid block. With overhung springs block under the spring saddle above the frame or take down spring and block end of equalizer above frame. All rods and eccentric straps must be removed on the broken side.

160. Should the back section of the side rod on a six-wheel connected engine break, what would you do?

A. Would take off both back sections of side rods and run in with forward wheels connected, taking whatever train the engine would handle.

161. Should the forward side rod section of the engine mentioned break, what must be done?

A. Take off all side rods and run light to end of division.

162. Should the back or front section of a side rod on a consolidation or other 8-wheel connected engine break, how would you disconnect?

A. Would take off both back or front connections, as the case might be, and run in whatever train the engine could handle.

163. Should the middle side rod connection of a consolidation engine break, how would you disconnect?

A. Would take down all side rods and run in without train.

164. Should the main rod crank pin break close to the wheel, what would you do?

A. Would take off the main rod on disabled side, block the cross-head and cover the steam ports. Then I would take down all the side rods and proceed light.

165. What would you do if the intermediate side rod was broken on a consolidation engine having the eccentrics on the axle ahead of the main wheel?

A. Would send to the nearest telegraph office, giving notice that my engine had to be towed in. Meanwhile I would take down the broken rod, but leave the others up.

166. In case the forward tire of a ten-wheel engine should break, what would you do?

A. Would jack up the wheel with the broken tire or run it upon a block until it was clear, off the rail and the box up in the driving box jaws. Then take out the oil cellar and make a block to fit between the journal and the pedestal brace to carry the disabled wheel center clear of the rail. Would then run in without disconnecting, provided the rod had not been damaged by broken tire. Should the front side rod section have been bent by the broken tire, would remove all rods and run in light.

167. Should the engine you are running break a main tire, what would be your procedure?

A. Would block up the axle wheel the thickness of the tire, slack up the main and side rod keys and run in carefully without train.

168. Should the back tire break, how would you manage?

A. Would block up the wheel and axle far enough to clear the rail, would take off the back sections of side rods and run in carefully, taking particular caution in passing over frogs and switches.

169. Should the back tire of an eight-wheel engine break, what would you do?

A. Would block up wheel as in other cases and would take down side rods. Would proceed slowly, but would not attempt to back up. Would consult with headquarters at nearest telegraph station.

170. What should be considered a bad tender or engine track wheel?

A. One with sharp flange, or flat or shelled-out spots in tread of wheel 2½ ins. or more in length.

171. What should be done if an engine truck wheel or axle breaks?

A. It should be entirely removed or blocked up so as to have the wheel clear of the rail, and the truck frame should be securely fastened to the engine frame by chains.

172. What should be done if a tender truck wheel or axle should break?

A. Would pursue the same course as with the broken engine truck wheel and fasten the truck frame with chains to the tender frame. Then would move slowly and cautiously to the nearest point where repairs could be made.

173. How should an engine be blocked for a broken engine truck spring or equalizer? Also for broken tender truck spring.

A. If pilot will not be too low, let truck frame ride on boxes; otherwise, block between top of boxes and truck frame. Blocking for a broken spring will vary according to the type of truck used. Some have a coil spring over each axle box and are easily taken care of; some have semi-elliptic springs with the spring

moderate rate of speed when, without this type of governor, the pumps would run at top speed. In running position of the brake valve, the spring chamber above the diaphragm portion of the excess pressure head is directly connected with the brake pipe, and any drop in brake pipe pressure will weaken the pressure that is assisting the regulating spring in holding the diaphragm valve to its seat, and if main reservoir pressure can remain at a figure 20 lbs. or more in excess of the falling brake pipe pressure, the pump or pumps will be stopped until the pressure above the diaphragm is restored to a figure within 20 lbs. of main reservoir pressure at which time the diaphragm valve will be seated and the pumps will be allowed to start For a complete demonstration of the action of this governor it is only necessary to pump up the air pressure on a locomotive and with the brake valve in running position, open an angle cock in the brake pipe and note the results. The pumps will not start until the pressure in the main reservoir falls to about 40 lbs., then they will work just fast enough to supply the volume of air escaping through the feed valve. If the pumps are started with the angle cock in the brake pipe open, the brake valve being in running position, about the same pressure will be compressed in the main reservoir and the pumps will run just fast enough to supply the volume of air that is escaping.

Naturally this is the only thing that can occur as the brake pipe is open and there is no air pressure above the diaphragms of the excess pressure head, and as soon as main reservoir pressure becomes strong enough to force the governor piston down and close the steam valve, the pumps will be stopped or controlled to the speed required.

Those who are inclined to jump at conclusions may have found considerable fault with this type of governor, but it has in the aggregate, saved many thousands of dollars in the purchase of air pump repair parts which will, from the standpoint of cost of maintenance justify its use, however, if some of the features sometimes thought to be unnecessary, are carefully considered, any cost of maintenance will be found to be of secondary importance.

In coupling to a train of uncharged cars, the proper position for charging the train is release position so that the fall of main reservoir pressure and the starting of the pumps will be prompt, and the governor cannot affect the operation of the pumps until after the desired brake pipe pressure is accumulated.

When the brake valve is returned to running position, the feed valve must maintain the brake pipe pressure against leakage or the pumps will be stopped by the action mentioned.

Now no matter how this action may be regarded, the fact remains that the engineer can turn this to a decided advantage to himself.

Under ordinary circumstances, he should never leave a terminal with his brake valve in release position unless descending a heavy grade should necessitate it, and if the train is charged in release position and the pumps are stopped every time the brake valve handle is brought to running position, proving that the escape of air from the brake pipe is in excess of the capacity of the feed valve, the leakage should be stopped, or if the fault is with the feed valve, it should be corrected before attempting to handle the train.

It is not necessary to open an angle cock to note the escape of air pressure in an effort to determine whether the feed valve is defective or whether the leakage is excessive. If the brake valve handle is placed on lap position until 90 or 100 lbs. is compressed in the main reservoir and then returned to the running position, the rate of fall in main reservoir pressure will show the condition of the feed valve so far as its ability to deliver the air pressure is concerned.

The rate of fall of main reservoir pressure through the feed valve will of course depend upon the volume and pressure in both the main reservoir, and the brake pipe into which the reservoir pressure is expanding, but with the modern main reservoir volume of from 60,000 to 80,000 cubic inches capacity, under about 100 lbs. air pressure the B 6 feed valve should reduce the main reservoir pressure by flowing into an empty or considerably depleted brake pipe at a rate of from about 30 to 40 lbs. per minute, that is, the main reservoir pressure should drop from 100 lbs. to 80 lbs., or from 90 lbs. to 70 lbs. in from 30 to 40 seconds.

If the pressure drops at approximately this rate it proves the feed valve to be in a condition that will enable it to supply any ordinary brake pipe leakage, and this refers to the B 6 or C 6 valve which should be used in present day service. It will be remembered that the size of compressor is not considered in this test, as it is only made under conditions where the governor stops the pump as soon as the brake valve is placed in running position. In this manner an observation of the gauge will show the condition of the brake pipe as well as the feed valve, and the train should not be moved from the terminal until it can be charged and the brake pipe pressure maintained with the brake valve handle in running position.

In order to do god braking in freight service the amount of brake pipe pressure to be exhausted by the initial reduction must be fixed upon, and it is determined by the length of train, type of brake equipments and track conditions, and if brake pipe leakage is sufficient to tax or exceed the capacity of the feed valve, the amount of brake pipe reduction cannot be regulated by the engineer, and he cannot be held responsible for results in stopping trains with excessive brake pipe leakage unless he leaves the terminal with the knowledge that the brake pipe leakage is excessive

The amount of brake pipe leakage a train is allowed to leave with often depends upon the capacity of the air pump, the 9½ or 11-inch pump, cannot supply air as fast as it can flow through the feed valve under a high pressure, consequently the leakage must necessarily be looked after in many instances before the train can be charged, but with the 8½ C. C., the New York No. 5 or two 11-inch pumps, a figure of brake pipe pressure can be reached against leakage that the feed valve cannot maintain, and if an engineer will leave a terminal point with the intention of running the brake valve in release position, he should expect to be held responsible for the safety of, and any possible damage to the train.

This governor top will not require any additional or added attention, but it should be kept clean and sensitive enough to prevent main reservoir pressure from falling to within 12 or 14 lbs. of the adjustment of the feed valve.

There is also the possibility of a piece of foreign matter lodging on the diaphragm valve seat of either governor top which would throttle or possibly stop the pump, and in the event of any stoppage of the pump the air gauge should always be observed, and if there is more than 20 lbs. difference between main reservoir and brake pipe pressure, it indicates that there has been a decrease in brake pipe pressure or an increase in main reservoir pressure, and that the pumps will start as soon as the pressures near an equilibrium.

If, however, the pump has stopped and the difference between main reservoir and brake pipe pressure is less than 20 lbs., it indicates that the governor is at fault, or that the pump has failed, due to some disorder in itself. It is assumed that the air pressure is at a figure somewhat below the adjustment of the maximum governor head, then if there is a blow or waste of air from the relief port of the governor, it proves the governor to be at fault, but if there is no indication that the governor has stopped the pump, we would at once ascertain whether the pump is receiving an ample supply of steam to run it.

If the engineer makes it his business to notice whether the relief port is open before leaving the engine house, he may avoid a bad half hour of pump or governor trouble along the road.

Should this excess pressure head become defective en route, it will be noticed either by a failure to stop the pump when the desired pressure is attained, in which event the maximum head will con-